



PULSELEDGER / PROGRAMME BLUEPRINT

Real-Time Financial Crime Intelligence

A human-centred, graph-aware data and AI platform
planned as a twelve-month portfolio programme.

75

PAGES

1M

EVENT TARGET

7

WORKSTREAMS

12

MONTHS

Long-term implementation and interview plan

Pablo Williams | Version 1.0 | 04 September 2026

EXECUTIVE DECISION

Build a decision platform, not a fraud dashboard

PulseLedger should prove that real-time data, machine learning, graph intelligence and controlled AI can improve investigator focus without hiding customer consequences.

PROGRAMME THESIS

Process a reproducible million-event stream, rank alerts within a fixed investigation budget, expose network evidence and support a human investigator through a fully traceable decision.

HUMAN PRINCIPLE

The interface starts with consequence, evidence and correction. It avoids alarmist colour, false certainty, performative dashboards and autonomous account restriction.

TECHNICAL PRINCIPLE

Every event, feature, score, graph signal, policy passage, proposal, approval and decision shares lineage. Replay and challenge are designed before optimisation.

PORTFOLIO OUTCOME

An interviewer can run the platform locally, reproduce the backtest, inspect the graph and workflow, review contracts and infrastructure, and challenge every CV number.

PRODUCT NORTH STAR

Help investigators find the few cases that deserve attention

The goal is not maximum alerts. It is the best customer-protecting allocation of constrained investigation capacity.

PRIMARY OUTCOME

Increase synthetic fraud-value capture at a fixed one-percent alert budget while reducing time spent assembling evidence.

GUARDRAILS

No autonomous account restriction; no hidden cohort trade-off; no untraceable feature or score; no model promotion without a fixed holdout and named risk acceptance.

EXPERIENCE STANDARD

A reviewer sees reason codes, network context, source time, uncertainty and next action in one coherent workspace. Every material decision remains editable before approval and contestable after.

SUCCESS HORIZON

A twelve-month programme ends with a controlled pilot decision, not a claim that synthetic performance guarantees banking outcomes.

ROADMAP AT A GLANCE

Twelve months of gated capability

Each quarter ends in a decision based on working evidence rather than architecture completion alone.

QUARTER 1 - FOUNDATION

Research investigators, define harm and metrics, build the event contract, generator, replay backbone, rules baseline and first human queue.

QUARTER 2 - INTELLIGENCE

Add time-safe features, tabular model, Neo4j domain, graph features, calibrated ensemble, evaluation harness and evidence explanations.

QUARTER 3 - CONTROLLED ASSISTANCE

Build the investigation agent, policy retrieval, MCP gateway, payload-bound approval, model registry, drift monitoring and failure exercises.

QUARTER 4 - PILOT READINESS

Run scale and adversarial tests, shadow champion/challenger models, rehearse incidents, validate benefits and package the production investment decision.

CONTENTS

Seventy-five pages, seven connected workstreams

The plan moves from human decisions to data, intelligence, controlled agency, operations, assurance and portfolio proof.

01

Human investigation product

Design around investigator judgment, customer consequence and supervisory accountability.

02

Streaming data foundation

Create a replayable, quality-controlled event backbone with clear ownership and lineage.

03

Detection and graph intelligence

Combine interpretable rules, tabular models and network signals under a fixed alert budget.

04

Investigation agent and workflow

Use AI to assemble evidence and draft decisions while keeping consequential authority human.

05

MLOps and platform engineering

Operate data, models and decisions as versioned services with measurable reliability and cost.

06

Security, regulation and resilience

Make customer protection, auditability and failure response enforceable rather than decorative.

07

Long-term delivery and portfolio proof

Sequence a credible year of delivery and turn each milestone into interview-verifiable evidence.

WORKSTREAM 01

Human investigation product

Design around investigator judgment, customer consequence and supervisory accountability.

01

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

01.1 Operating problem and mandate

Define the financial-crime decision, affected people, sponsor, alert budget and evidence needed to fund a pilot.

SIGNED MANDATE AND DECISION LOG**01.2 Investigator field research**

Observe triage, evidence gathering, queue pressure, handoffs, overrides and supervisory review.

RESEARCH SYNTHESIS AND OPPORTUNITY MAP**01.3 Customer consequence mapping**

Trace false positives, delayed payments, account restrictions and vulnerable-customer impacts.

HARM-AWARE SERVICE BLUEPRINT**01.4 Jobs and decision moments**

Model investigator, supervisor, fraud strategy, operations, compliance and customer-support needs.

PRIORITISED JOBS-TO-BE-DONE**01.5 Alert triage experience**

Design a queue that surfaces exposure, confidence, reason, age, SLA and network context without dark patterns.

ACCESSIBLE QUEUE SPECIFICATION**01.6 Investigation workspace**

Unify transaction timeline, customer profile, device evidence, graph links, policies, notes and decisions.

SCREEN AND INTERACTION CONTRACT**01.7 Human approval and contestability**

Require meaningful review for payment holds and restrictions; preserve correction and appeal routes.

APPROVAL AND CUSTOMER-REDRESS FLOWS**01.8 Adoption and capability**

Train calibrated trust, evidence reading, override reasons, model limitations and incident escalation.

ROLE-BASED LEARNING PROGRAMME

MODULE 01.1

Operating problem and mandate

Define the financial-crime decision, affected people, sponsor, alert budget and evidence needed to fund a pilot.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for operating problem and mandate: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Signed mandate and decision log is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed operating problem and mandate as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 01.2

Investigator field research

Observe triage, evidence gathering, queue pressure, handoffs, overrides and supervisory review.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for investigator field research: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Research synthesis and opportunity map is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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I designed investigator field research as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 01.3

Customer consequence mapping

Trace false positives, delayed payments, account restrictions and vulnerable-customer impacts.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for customer consequence mapping: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Harm-aware service blueprint is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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I designed customer consequence mapping as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 01.4

Jobs and decision moments

Model investigator, supervisor, fraud strategy, operations, compliance and customer-support needs.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for jobs and decision moments: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Prioritised jobs-to-be-done is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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I designed jobs and decision moments as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 01.5

Alert triage experience

Design a queue that surfaces exposure, confidence, reason, age, SLA and network context without dark patterns.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for alert triage experience: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Accessible queue specification is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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MODULE 01.6

Investigation workspace

Unify transaction timeline, customer profile, device evidence, graph links, policies, notes and decisions.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for investigation workspace: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Screen and interaction contract is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

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MODULE 01.7

Human approval and contestability

Require meaningful review for payment holds and restrictions; preserve correction and appeal routes.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for human approval and contestability: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Approval and customer-redress flows is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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I designed human approval and contestability as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 01.8

Adoption and capability

Train calibrated trust, evidence reading, override reasons, model limitations and incident escalation.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for adoption and capability: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Role-based learning programme is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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INTERVIEW DEFENCE

I designed adoption and capability as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

WORKSTREAM 02

Streaming data foundation

Create a replayable, quality-controlled event backbone with clear ownership and lineage.

02

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

02.1 Canonical payment event

Version identifiers, parties, device, location, amount, merchant, authentication and event-time fields.

SCHEMA REGISTRY CONTRACT

02.2 Synthetic event generator

Produce one million reproducible events across ordinary behaviour, account takeover, mule activity and card testing.

FIXED-SEED STREAMING GENERATOR

02.3 Kafka and topic design

Separate raw, validated, enriched, scored, alert and decision topics with retention and ownership.

TOPIC AND PARTITION STRATEGY

02.4 Event-time correctness

Specify watermarks, lateness, ordering, duplicate handling and deterministic replay.

EVENT-TIME TEST SUITE

02.5 Real-time feature computation

Calculate velocity, novelty, distance, recipient risk, device sharing and failed-payment windows.

ONLINE FEATURE CONTRACTS

02.6 Offline lakehouse design

Land immutable history, curated outcomes and training snapshots with reproducible time travel.

BRONZE-SILVER-GOLD DATA PRODUCTS

02.7 Data quality and observability

Monitor freshness, completeness, uniqueness, validity, distribution and cross-topic reconciliation.

DATA SLOS AND INCIDENT RULES

02.8 Privacy and retention

Minimise sensitive fields, tokenise identifiers, define purpose, deletion and investigation holds.

DPIA AND RETENTION SCHEDULE

MODULE 02.1

Canonical payment event

Version identifiers, parties, device, location, amount, merchant, authentication and event-time fields.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for canonical payment event: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Schema registry contract is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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DELIVERY OWNERSHIP

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INTERVIEW DEFENCE

I designed canonical payment event as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.2

Synthetic event generator

Produce one million reproducible events across ordinary behaviour, account takeover, mule activity and card testing.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for synthetic event generator: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Fixed-seed streaming generator is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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INTERVIEW DEFENCE

I designed synthetic event generator as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.3

Kafka and topic design

Separate raw, validated, enriched, scored, alert and decision topics with retention and ownership.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for kafka and topic design: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Topic and partition strategy is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

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INTERVIEW DEFENCE

I designed kafka and topic design as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.4

Event-time correctness

Specify watermarks, lateness, ordering, duplicate handling and deterministic replay.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for event-time correctness: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Event-time test suite is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

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INTERVIEW DEFENCE

I designed event-time correctness as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.5

Real-time feature computation

Calculate velocity, novelty, distance, recipient risk, device sharing and failed-payment windows.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for real-time feature computation: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Online feature contracts is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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I designed real-time feature computation as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.6

Offline lakehouse design

Land immutable history, curated outcomes and training snapshots with reproducible time travel.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for offline lakehouse design: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Bronze-silver-gold data products is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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INTERVIEW DEFENCE

I designed offline lakehouse design as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.7

Data quality and observability

Monitor freshness, completeness, uniqueness, validity, distribution and cross-topic reconciliation.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for data quality and observability: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Data SLOs and incident rules is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

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I designed data quality and observability as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 02.8

Privacy and retention

Minimise sensitive fields, tokenise identifiers, define purpose, deletion and investigation holds.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for privacy and retention: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- DPIA and retention schedule is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

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I designed privacy and retention as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

WORKSTREAM 03

Detection and graph intelligence

Combine interpretable rules, tabular models and network signals under a fixed alert budget.

03

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

03.1 Detection strategy

Define threats, decision latency, cost asymmetry, alert capacity and model-versus-rule ownership.

DETECTION COVERAGE MATRIX

03.2 Rules baseline

Implement transparent velocity, value, geo, device and recipient rules as the minimum benchmark.

VERSIONED RULES SERVICE

03.3 Tabular risk model

Train an interpretable baseline using leakage-safe historical features and calibrated probabilities.

MODEL ARTIFACT AND CARD

03.4 Graph domain model

Represent customers, accounts, devices, IPs, merchants, beneficiaries and transactions.

NEO4J SCHEMA AND CONSTRAINTS

03.5 Graph risk features

Calculate shared-device risk, risky-neighbour ratio, cycles, fan-in, fan-out and shortest paths.

ONLINE GRAPH FEATURE CONTRACT

03.6 Ensemble and thresholds

Blend rule, tabular and graph evidence and choose thresholds by alert capacity and exposure.

CHAMPION SCORING POLICY

03.7 Explainability and evidence

Translate contributions into decision evidence without presenting correlation as certainty.

REASON-CODE AND EXPLANATION STANDARD

03.8 Backtest and ablation

Compare rules, tabular and graph-augmented systems on the same one-million-event holdout.

REPRODUCIBLE EVALUATION REPORT

MODULE 03.1

Detection strategy

Define threats, decision latency, cost asymmetry, alert capacity and model-versus-rule ownership.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for detection strategy: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Detection coverage matrix is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed detection strategy as part of the operating system, not an isolated feature.
It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.2

Rules baseline

Implement transparent velocity, value, geo, device and recipient rules as the minimum benchmark.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for rules baseline: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Versioned rules service is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed rules baseline as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.3

Tabular risk model

Train an interpretable baseline using leakage-safe historical features and calibrated probabilities.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for tabular risk model: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Model artifact and card is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed tabular risk model as part of the operating system, not an isolated feature.
It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.4

Graph domain model

Represent customers, accounts, devices, IPs, merchants, beneficiaries and transactions.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for graph domain model: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Neo4j schema and constraints is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed graph domain model as part of the operating system, not an isolated feature.
It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.5

Graph risk features

Calculate shared-device risk, risky-neighbour ratio, cycles, fan-in, fan-out and shortest paths.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for graph risk features: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Online graph feature contract is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed graph risk features as part of the operating system, not an isolated feature.
It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.6

Ensemble and thresholds

Blend rule, tabular and graph evidence and choose thresholds by alert capacity and exposure.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for ensemble and thresholds: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Champion scoring policy is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed ensemble and thresholds as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.7

Explainability and evidence

Translate contributions into decision evidence without presenting correlation as certainty.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for explainability and evidence: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Reason-code and explanation standard is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed explainability and evidence as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 03.8

Backtest and ablation

Compare rules, tabular and graph-augmented systems on the same one-million-event holdout.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for backtest and ablation: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Reproducible evaluation report is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed backtest and ablation as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

WORKSTREAM 04

Investigation agent and workflow

Use AI to assemble evidence and draft decisions while keeping consequential authority human.

04

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

04.1 Agent purpose and boundaries

Permit search, summarisation, timeline assembly and drafting; prohibit autonomous account restriction.

SYSTEM CARD AND AUTHORITY MATRIX

04.2 Investigation state machine

Model open, enrich, retrieve, analyse, propose, approve, execute, notify and close states.

WORKFLOW TRANSITION SPECIFICATION

04.3 Policy retrieval

Index dated fraud, customer-support and vulnerable-customer procedures with owner and jurisdiction metadata.

GROUNDING RETRIEVAL SERVICE

04.4 Tool gateway and MCP

Expose narrow read, propose and execute tools with strict schemas, scoped identity and audit context.

MCP-COMPATIBLE TOOL CONTRACTS

04.5 Payload-bound approval

Bind restriction approval to customer, action, scope, reason, expiry, approver and immutable hash.

APPROVAL AND REPLAY CONTROLS

04.6 Prompt-injection defence

Treat events and documents as untrusted data; apply content boundaries and default-deny tools.

ADVERSARIAL EVALUATION SUITE

04.7 Investigation memory

Keep case memory scoped, redacted, source-linked and deleted according to retention policy.

MEMORY LIFECYCLE DESIGN

04.8 Quality and human oversight

Measure grounding, completeness, harmful omission, override, time saved and reviewer calibration.

AGENT EVALUATION SCORECARD

MODULE 04.1

Agent purpose and boundaries

Permit search, summarisation, timeline assembly and drafting; prohibit autonomous account restriction.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for agent purpose and boundaries: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- System card and authority matrix is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed agent purpose and boundaries as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.2

Investigation state machine

Model open, enrich, retrieve, analyse, propose, approve, execute, notify and close states.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for investigation state machine: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Workflow transition specification is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed investigation state machine as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.3

Policy retrieval

Index dated fraud, customer-support and vulnerable-customer procedures with owner and jurisdiction metadata.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for policy retrieval: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Grounded retrieval service is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed policy retrieval as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.4

Tool gateway and MCP

Expose narrow read, propose and execute tools with strict schemas, scoped identity and audit context.

DETAILED DELIVERY SEQUENCE

- 1

Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2

Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3

Define the contract for tool gateway and mcp: inputs, outputs, state, ownership, quality rules, version and retention.
- 4

Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5

Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6

Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

MCP-compatible tool contracts is versioned and linked to its source evidence.

Definitions, exclusions, failure behaviour and decision threshold are explicit.

A named operational owner accepts maintenance and incident responsibility.

Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT Outcome and adoption	ENGINEERING Service and failure	CONTROL Risk acceptance	EVIDENCE Metric and lineage
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INTERVIEW DEFENCE

I designed tool gateway and mcp as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.5

Payload-bound approval

Bind restriction approval to customer, action, scope, reason, expiry, approver and immutable hash.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for payload-bound approval: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Approval and replay controls is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed payload-bound approval as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.6

Prompt-injection defence

Treat events and documents as untrusted data; apply content boundaries and default-deny tools.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for prompt-injection defence: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Adversarial evaluation suite is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed prompt-injection defence as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.7

Investigation memory

Keep case memory scoped, redacted, source-linked and deleted according to retention policy.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for investigation memory: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Memory lifecycle design is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed investigation memory as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 04.8

Quality and human oversight

Measure grounding, completeness, harmful omission, override, time saved and reviewer calibration.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for quality and human oversight: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Agent evaluation scorecard is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed quality and human oversight as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

WORKSTREAM 05

MLOps and platform engineering

Operate data, models and decisions as versioned services with measurable reliability and cost.

05

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

05.1 Model registry and lineage

Link code, data snapshot, features, model, threshold, evaluation, approval and deployment.

RELEASE EVIDENCE BUNDLE

05.2 Champion-challenger delivery

Score challengers silently, compare outcomes and promote only through evidence gates.

SHADOW DEPLOYMENT PIPELINE

05.3 Feature parity

Prevent training-serving skew through shared definitions, point-in-time joins and parity tests.

FEATURE PARITY GATE

05.4 Drift and performance monitoring

Detect input, feature, score, outcome and calibration drift by relevant cohorts.

MONITORING AND RESPONSE PLAYBOOK

05.5 Service-level objectives

Set throughput, availability, end-to-end latency, event lag, recovery and decision trace targets.

SLO AND ERROR-BUDGET POLICY

05.6 Cloud and Kubernetes architecture

Map local services to AWS streaming, compute, graph, storage, model and observability capabilities.

TARGET DEPLOYMENT ARCHITECTURE

05.7 Infrastructure and release automation

Use Terraform, containers, signed builds, policy checks, environment promotion and rollback.

CI/CD AND IAC REPOSITORY

05.8 FinOps and capacity engineering

Measure cost per million events, per alert and per prevented loss with load forecasts and budgets.

CAPACITY AND UNIT-COST MODEL

MODULE 05.1

Model registry and lineage

Link code, data snapshot, features, model, threshold, evaluation, approval and deployment.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for model registry and lineage: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Release evidence bundle is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed model registry and lineage as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.2

Champion-challenger delivery

Score challengers silently, compare outcomes and promote only through evidence gates.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for champion-challenger delivery: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Shadow deployment pipeline is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed champion-challenger delivery as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.3

Feature parity

Prevent training-serving skew through shared definitions, point-in-time joins and parity tests.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for feature parity: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Feature parity gate is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed feature parity as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.4

Drift and performance monitoring

Detect input, feature, score, outcome and calibration drift by relevant cohorts.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for drift and performance monitoring: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Monitoring and response playbook is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed drift and performance monitoring as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.5

Service-level objectives

Set throughput, availability, end-to-end latency, event lag, recovery and decision trace targets.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for service-level objectives: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- SLO and error-budget policy is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed service-level objectives as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.6

Cloud and Kubernetes architecture

Map local services to AWS streaming, compute, graph, storage, model and observability capabilities.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for cloud and kubernetes architecture: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Target deployment architecture is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed cloud and kubernetes architecture as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.7

Infrastructure and release automation

Use Terraform, containers, signed builds, policy checks, environment promotion and rollback.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for infrastructure and release automation: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- CI/CD and IaC repository is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed infrastructure and release automation as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 05.8

FinOps and capacity engineering

Measure cost per million events, per alert and per prevented loss with load forecasts and budgets.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for finops and capacity engineering: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Capacity and unit-cost model is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed finops and capacity engineering as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

WORKSTREAM 06

Security, regulation and resilience

Make customer protection, auditability and failure response enforceable rather than decorative.

06

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

06.1 Threat and abuse model

Cover event poisoning, identity abuse, model evasion, graph poisoning, prompt injection and insider misuse.

THREAT MODEL AND MITIGATIONS**06.2 Identity and least privilege**

Propagate user, service, tenant, case and purpose attributes across every read and action.

AUTHORISATION POLICY TESTS**06.3 Encryption and secrets**

Protect events, features, graphs, cases, models, logs and backups with rotation and separation of duties.

KEY AND SECRET LIFECYCLE**06.4 Audit and decision lineage**

Reconstruct who knew what, which model ran, what evidence existed and why an action occurred.

IMMUTABLE AUDIT CONTRACT**06.5 Fairness and customer outcomes**

Measure friction, false positives, delay and restriction outcomes across defensible cohorts.

OUTCOME MONITORING STANDARD**06.6 Resilience and degradation**

Design replay, deduplication, circuit breaking, queue back-pressure, manual fallback and safe model bypass.

FAILURE-MODE CATALOGUE**06.7 Incident response and kill switch**

Define severity, authority, containment, customer correction, regulatory notification and recovery.

EXERCISED INCIDENT RUNBOOKS**06.8 Assurance and governance**

Use design authority, model risk, data governance, security and business owners with recorded decisions.

GOVERNANCE CALENDAR AND RACI

MODULE 06.1

Threat and abuse model

Cover event poisoning, identity abuse, model evasion, graph poisoning, prompt injection and insider misuse.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for threat and abuse model: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Threat model and mitigations is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed threat and abuse model as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.2

Identity and least privilege

Propagate user, service, tenant, case and purpose attributes across every read and action.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for identity and least privilege: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Authorisation policy tests is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed identity and least privilege as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.3

Encryption and secrets

Protect events, features, graphs, cases, models, logs and backups with rotation and separation of duties.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for encryption and secrets: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Key and secret lifecycle is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed encryption and secrets as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.4

Audit and decision lineage

Reconstruct who knew what, which model ran, what evidence existed and why an action occurred.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for audit and decision lineage: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Immutable audit contract is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed audit and decision lineage as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.5

Fairness and customer outcomes

Measure friction, false positives, delay and restriction outcomes across defensible cohorts.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for fairness and customer outcomes: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Outcome monitoring standard is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed fairness and customer outcomes as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.6

Resilience and degradation

Design replay, deduplication, circuit breaking, queue back-pressure, manual fallback and safe model bypass.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for resilience and degradation: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Failure-mode catalogue is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed resilience and degradation as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.7

Incident response and kill switch

Define severity, authority, containment, customer correction, regulatory notification and recovery.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for incident response and kill switch: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Exercised incident runbooks is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed incident response and kill switch as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 06.8

Assurance and governance

Use design authority, model risk, data governance, security and business owners with recorded decisions.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for assurance and governance: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Governance calendar and RACI is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed assurance and governance as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

WORKSTREAM 07

Long-term delivery and portfolio proof

Sequence a credible year of delivery and turn each milestone into interview-verifiable evidence.

07

EXIT CONDITION

Named owners can make the next delivery decision using accepted evidence, explicit residual risk and a rehearsed fallback.

07.1 Twelve-month roadmap

Stage foundation, proof, shadow, controlled pilot, scale and optimisation around evidence gates.

INTEGRATED PROGRAMME PLAN**07.2 Team topology and ownership**

Define product, investigation, streaming, ML, graph, platform, security and design responsibilities.

TEAM CHARTER AND DEPENDENCY MAP**07.3 Backlog and architecture runway**

Prioritise vertical outcomes while funding schema, observability, security and replay capabilities early.

OUTCOME-LED BACKLOG**07.4 Testing and quality strategy**

Cover domain, schema, stream, feature, model, graph, agent, security, performance and recovery layers.

RISK-BASED TEST STRATEGY**07.5 Pilot and benefits realisation**

Pre-register metrics, run matched queues, monitor customer outcomes and validate cashable value.

PILOT PROTOCOL AND BENEFITS LEDGER**07.6 Open-source portfolio packaging**

Publish safe synthetic assets, reproducible commands, diagrams, ADRs, runbooks and claim boundaries.

INTERVIEW-READY REPOSITORY**07.7 Demonstration and defence**

Tell a 25-minute decision story and prepare challenges on data, models, graphs, controls and economics.

DEMO SCRIPT AND QUESTION BANK**07.8 CV evidence ledger**

Attach every scale, quality, performance and value number to a command, artifact and truthful boundary.

AUDITABLE CV CLAIM REGISTER

MODULE 07.1

Twelve-month roadmap

Stage foundation, proof, shadow, controlled pilot, scale and optimisation around evidence gates.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for twelve-month roadmap: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Integrated programme plan is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed twelve-month roadmap as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 07.2

Team topology and ownership

Define product, investigation, streaming, ML, graph, platform, security and design responsibilities.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for team topology and ownership: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Team charter and dependency map is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

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Outcome and adoption

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Service and failure

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Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed team topology and ownership as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 07.3

Backlog and architecture runway

Prioritise vertical outcomes while funding schema, observability, security and replay capabilities early.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for backlog and architecture runway: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Outcome-led backlog is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed backlog and architecture runway as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 07.4

Testing and quality strategy

Cover domain, schema, stream, feature, model, graph, agent, security, performance and recovery layers.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for testing and quality strategy: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Risk-based test strategy is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

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EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed testing and quality strategy as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 07.5

Pilot and benefits realisation

Pre-register metrics, run matched queues, monitor customer outcomes and validate cashable value.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for pilot and benefits realisation: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Pilot protocol and benefits ledger is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

ENGINEERING

Service and failure

CONTROL

Risk acceptance

EVIDENCE

Metric and lineage

INTERVIEW DEFENCE

I designed pilot and benefits realisation as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 07.6

Open-source portfolio packaging

Publish safe synthetic assets, reproducible commands, diagrams, ADRs, runbooks and claim boundaries.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for open-source portfolio packaging: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Interview-ready repository is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

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INTERVIEW DEFENCE

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MODULE 07.7

Demonstration and defence

Tell a 25-minute decision story and prepare challenges on data, models, graphs, controls and economics.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for demonstration and defence: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Demo script and question bank is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

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Metric and lineage

INTERVIEW DEFENCE

I designed demonstration and defence as part of the operating system, not an isolated feature. It has a human decision, typed contract, owner, measurable gate and safe failure path.

MODULE 07.8

CV evidence ledger

Attach every scale, quality, performance and value number to a command, artifact and truthful boundary.

DETAILED DELIVERY SEQUENCE

- 1 Name the investigator or customer decision this module changes, its accountable owner and the consequence of delay or error.
- 2 Collect direct evidence from people, event data, existing controls, policies and system behaviour; keep hypotheses visibly separate.
- 3 Define the contract for cv evidence ledger: inputs, outputs, state, ownership, quality rules, version and retention.
- 4 Prototype the happy path, ambiguous path, harmful false positive, infrastructure failure and deliberate abuse path.
- 5 Connect the output to streaming, graph, model, case, audit and benefits evidence through stable identifiers and trace context.
- 6 Review with an investigator, control owner and engineer; capture disagreement and decide accept, revise, defer or stop.

ACCEPTANCE EVIDENCE

- Auditable CV claim register is versioned and linked to its source evidence.
- Definitions, exclusions, failure behaviour and decision threshold are explicit.
- A named operational owner accepts maintenance and incident responsibility.
- Automated checks or a documented review verify the material requirement.

HUMAN-QUALITY TEST

Can a busy investigator understand the evidence, consequence and available correction without learning the internal implementation? Can a supervisor reconstruct and challenge the decision later?

DELIVERY OWNERSHIP

PRODUCT

Outcome and adoption

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INTERVIEW DEFENCE

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It has a human decision, typed contract, owner, measurable gate and safe failure path.

APPENDIX A

Reference technology map

Local tools preserve reproducibility; target services show enterprise judgement.

LOCAL PROOF

- Next.js and TypeScript investigator experience
- Python streaming generator and evaluation harness
- Redpanda Kafka-compatible event backbone
- PostgreSQL cases and outcomes
- Neo4j account-device-payment graph
- OpenTelemetry-compatible trace contracts

AWS TARGET

- MSK or Kinesis for managed streaming
- Flink or Lambda for stateful enrichment
- S3 and Iceberg for historical data products
- SageMaker or container model serving
- Neptune or Neo4j Aura for graph intelligence
- EKS, ECS or Lambda chosen by workload evidence

APPENDIX B

Metric and decision dictionary

Metrics are tied to a fixed alert capacity and customer outcomes.

DETECTION

- Precision and recall at one-percent alert budget
- Fraud value captured, not only fraud count
- False positives per 1,000 transactions
- Calibration error and threshold stability
- Graph uplift over tabular baseline
- Time-to-detection by fraud pattern

PLATFORM AND PEOPLE

- P95 scoring latency and event lag
- Sustained events per second and recovery time
- Investigator evidence-assembly minutes
- Override and escalation quality
- Customer restriction and delay outcomes
- Cost per million events and per useful alert

APPENDIX C

Repository architecture

A public repository that a senior engineer can navigate quickly.

RUNTIME

- app and components - investigator product
- lib - domain, scoring, graph and workflow logic
- app/api - resource and action endpoints
- analytics - generator, backtest and validator
- data - aggregates and safe demonstration samples
- contracts - AsyncAPI, OpenAPI and MCP schemas

DELIVERY

- infra - Terraform and deployment notes
- docs/adr - material architecture decisions
- docs/runbooks - operational response
- tests - domain, safety and contract checks
- docker-compose.yml - optional local dependencies
- GitHub Actions - reproducible quality gates

APPENDIX D

Risk and control starter

The first assurance conversation begins before model selection.

CUSTOMER AND MODEL

- False positive restriction or delay
- Missed fraud and financial loss
- Unequal friction across customer cohorts
- Misleading explanation or false certainty
- Outcome label delay and feedback bias
- Concept drift and attacker adaptation

PLATFORM AND AGENT

- Event or graph poisoning
- Training-serving skew
- Prompt injection and data exfiltration
- Excessive tool authority
- Approval bait-and-switch or replay
- Duplicate actions during retry and recovery

APPENDIX E

Backtest protocol

A reproducible synthetic evaluation that is difficult to misrepresent.

DESIGN

- One million streamed transactions
- Fixed seed and published generator
- Same holdout for rules, tabular and graph ensemble
- One-percent alert capacity constraint
- Ablation and pattern-level cohorts
- Bootstrap or Wilson uncertainty where appropriate

CLAIM BOUNDARY

- All people, institutions and transactions are fictional
- Scores and outcomes are simulated
- No randomised or production experiment is implied
- Synthetic latency is not cloud load evidence
- Value capture is not recovered cash
- Every CV statement retains the boundary

APPENDIX F

Twenty-five minute demonstration

A deliberate interview story from operational problem to controlled decision.

NARRATIVE

- 3 min - mandate, alert budget and human consequence
- 4 min - live stream and data quality
- 5 min - alert queue and network investigation
- 4 min - agent evidence and approval boundary
- 5 min - model operations and backtest
- 4 min - architecture, risks and next investment gate

CHALLENGE - PROMPTS

- Why graph rather than more tabular features?
- How do you prevent training-serving leakage?
- What happens when Kafka or graph lookup fails?
- How do you set a threshold with limited investigators?
- How can a customer correct a false positive?
- Which result would make you stop the pilot?

APPENDIX G

Completion and CV evidence checklist

The repository is ready only when code, evidence and wording agree.

RELEASE READINESS

- Fresh checkout runs without paid services
- Million-event backtest reproduces the aggregate
- All tests, types, lint and build pass locally and in CI
- Routine, ambiguous and blocked workflows work
- No secret or real personal data is committed
- Architecture and contracts match runtime behaviour

CV READINESS

- Every number links to an artifact and command
- Synthetic and simulated remain in outcome claims
- Scale claim states generated or processed events
- Latency claim identifies local test conditions
- Business value is modelled, not delivered
- Limitations are easy to find in README and UI